REPORT

Zajęcia: Analog and digital electronic circuits
Teacher: prof. dr hab. Vasyl Martsenyuk

Lab 1
Date 27.02.26
Topic: "Windowing in Spectral Analysis (FFT)"
Variant 13

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stacjonarne,
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Gr 1.a

1. Problem statement:

Simulate two random signals with different power levels. Analyze and compare their energy and standard deviation. Estimate and compare their PSDs using Welch with Hamming and Hann windows.

2. Input data:

Variant 13:

fs = 1000 # sampling frequency
N = 4096 # number of samples

3. Commands used (or GUI):

1. Import libraries

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import welch
```

2. Simulation parameters

```
# PARAMETERS
fs = 1000          # sampling frequency
N = 4096           # number of samples
```

3. Calculation of energy and standard deviation

Signal energy:

$$E = \sum_{n=0}^{N-1} x[n]^2$$

```
# ENERGY AND STD
energy1 = np.sum(signal1**2)
energy2 = np.sum(signal2**2)

std1 = np.std(signal1)
std2 = np.std(signal2)

print("----- SIGNAL 1 -----")
print("Energy:", energy1)
print("Standard deviation:", std1)

print("\n----- SIGNAL 2 -----")
print("Energy:", energy2)
print("Standard deviation:", std2)
```

Output:

----- SIGNAL 1 -----
Energy: 3962.4935759474038
Standard deviation: 0.9832072871275006

----- SIGNAL 2 -----
Energy: 36228.507340144795
Standard deviation: 2.9739950741482555

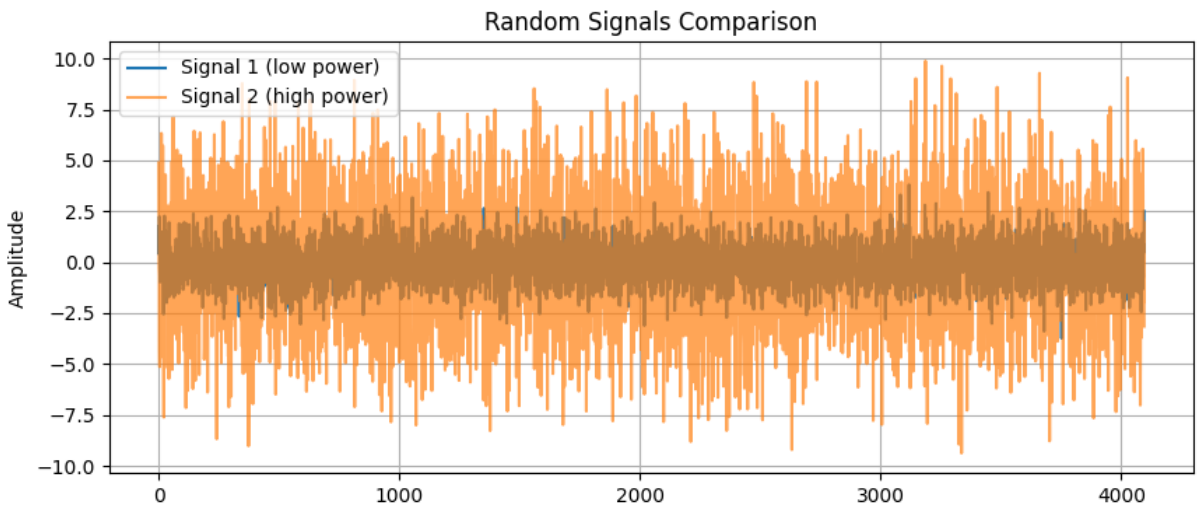
Conclusion:

Signal 2 has a higher standard deviation, so it has more power and energy.

4. Signal graph

```
# PLOT SIGNALS
plt.figure(figsize=(10,4))
plt.plot(signal1, label="Signal 1 (low power)")
plt.plot(signal2, label="Signal 2 (high power)", alpha=0.7)
plt.title("Random Signals Comparison")
plt.xlabel("Sample")
plt.ylabel("Amplitude")
plt.legend()
plt.grid(True)
plt.show()
```

Output:

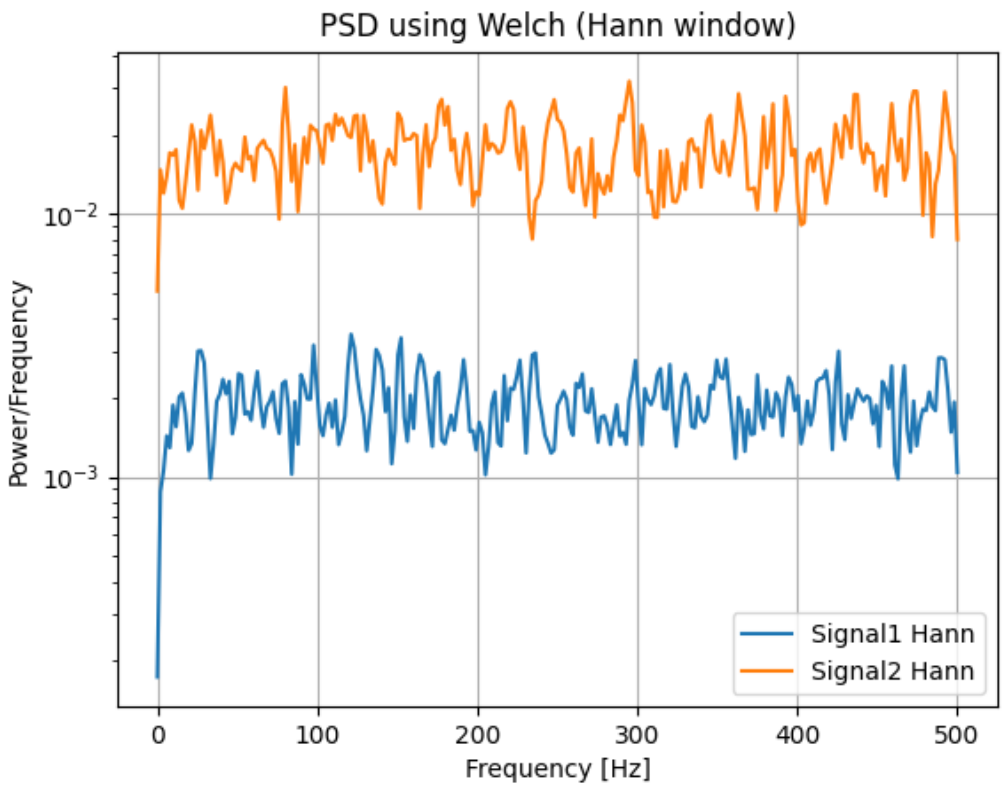


5. Welch-Hann window PSD

```
# PSD - WELCH (HANN WINDOW)
f1_hann, Pxx1_hann = welch(signal1, fs=fs, window='hann', nperseg=512)
f2_hann, Pxx2_hann = welch(signal2, fs=fs, window='hann', nperseg=512)

plt.figure()
plt.semilogy(f1_hann, Pxx1_hann, label="Signal1 Hann")
plt.semilogy(f2_hann, Pxx2_hann, label="Signal2 Hann")
plt.title("PSD using Welch (Hann window)")
plt.xlabel("Frequency [Hz]")
plt.ylabel("Power/Frequency")
plt.legend()
plt.grid(True)
plt.show()
```

Output:

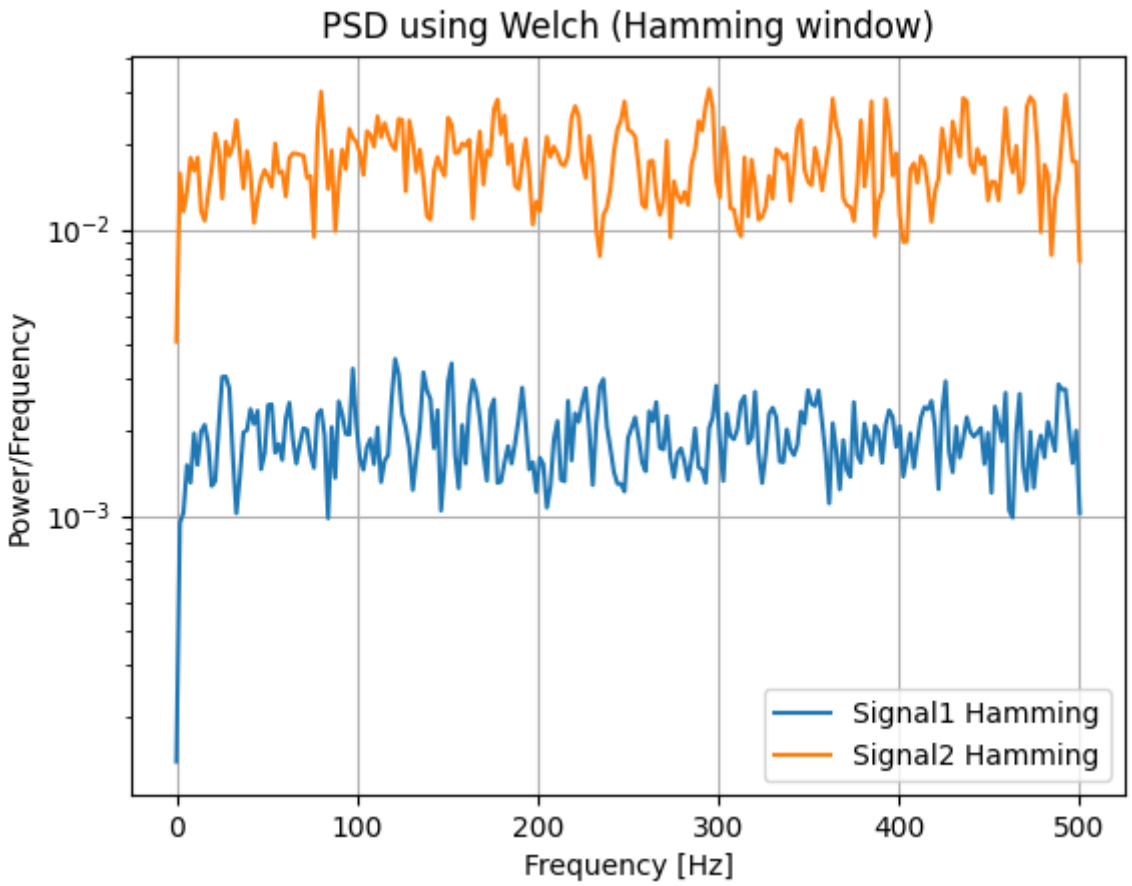


6. Welch-window Hamming PSD

```
# PSD - WELCH (HAMMING WINDOW)
f1_hamming, Pxx1_hamming = welch(signal1, fs=fs, window='hamming',
nperseg=512)
f2_hamming, Pxx2_hamming = welch(signal2, fs=fs, window='hamming',
nperseg=512)

plt.figure()
plt.semilogy(f1_hamming, Pxx1_hamming, label="Signal1 Hamming")
plt.semilogy(f2_hamming, Pxx2_hamming, label="Signal2 Hamming")
plt.title("PSD using Welch (Hamming window)")
plt.xlabel("Frequency [Hz]")
plt.ylabel("Power/Frequency")
plt.legend()
plt.grid(True)
plt.show()
```

Output:



Student assignment (deliverables):

Submit a short report (1–3 pages) answering:

1. Explain spectral leakage in your own words. Why does it happen?

   Answer: Spectral leakage occurs because in an FFT we analyze only a finite-length segment of a signal. Truncating a signal abruptly (equivalent to multiplying it by a rectangular window) introduces discontinuities at the edges. In the frequency domain, multiplication in time corresponds to convolution, so the FFT of a truncated sinusoid spreads its energy into neighboring frequency bins instead of being confined to a single bin.

2. For a **strong+weak** tone mixture:

   Which window helps detect the weak tone best?
   Answer: The Blackman or Kaiser (β≈12) window helps detect the weak tone best because it has very low sidelobes.

   What is the trade-off you pay for that choice?
   Answer: The trade-off is a wider main-lobe, which reduces frequency resolution and makes closely spaced tones harder to separate.

3. For **two close tones of similar amplitude**:

   Which window gives best separation?
   Answer: The Rectangular window gives the best separation of close tones due to its narrowest main-lobe.

   How does changing N affect your conclusion?
   Answer: Increasing N decreases df = fs/N, narrowing main-lobes in Hz and improving true frequency resolution, making peaks easier to distinguish even for wider windows.

4. Provide at least two plots that support your answers (with labels and dB scale).

   Three close tones – full spectrum (dB): Compare Rectangular, Hann, Blackman, Kaiser. Label axes: Frequency [Hz], Magnitude [dB].

   Zoom around the peaks: Highlight how Rectangular resolves all three peaks while Blackman/Kaiser may merge them.

Conclusion:

Due to this lab, we learned how windowing affects spectral analysis with the FFT and how spectral leakage arises from finite-length signals. We observed that choosing the right window depends on the goal: low-sidelobe windows like Blackman or Kaiser help detect weak tones near strong ones, while Rectangular windows provide the narrowest main-lobe for resolving closely spaced frequencies. We also saw that increasing the FFT length N improves frequency resolution, highlighting the trade-off between leakage suppression and frequency clarity in practical signal analysis.

Link to remote repozytorium (e.g. GitHub):

https://github.com/KostiaGrekov/DSPja/tree/main/DSPja%202